Computational Statistics and Data Analysis 120 (2018) 25–41

Contents lists available at ScienceDirect

Computational Statistics and Data Analysis

journal homepage: www.elsevier.com/locate/csda

Fast symmetric additive covariance smoothing

Jona Cederbaum, Fabian Scheipl, Sonja Greven *

Department of Statistics, Faculty of Mathematics, Computer Science and Statistics, LMU Munich, Germany

ARTICLE INFO

Article history:
Received 22 September 2016
Received in revised form 2 November 2017
Accepted 3 November 2017
Available online 20 November 2017

Keywords:
Functional data
Longitudinal data
Functional principal components
Penalized splines

ABSTRACT

A fast bivariate smoothing approach for symmetric surfaces is proposed that has a wide range of applications. It is shown how it can be applied to estimate the covariance function in longitudinal data as well as multiple additive covariances in functional data with complex correlation structures. The proposed symmetric smoother can handle (possibly noisy) data sampled on a common, dense grid as well as irregularly or sparsely sampled data. Estimation is based on bivariate penalized spline smoothing using a mixed model representation and the symmetry is used to reduce computation time compared to the usual non-symmetric smoothers. The application of the approach in functional principal component analysis for very general functional linear mixed models is outlined and its practical value is demonstrated in two applications. The approach is evaluated in extensive simulations. Documented open source software is provided that implements the fast symmetric bivariate smoother building on established algorithms for additive models.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

Covariance functions play a central role in many areas of statistics. They summarize the dependency between stochastic observations and encode smoothness assumptions about (observed or latent) random processes. We propose a fast bivariate smoothing approach for symmetric surfaces which can estimate covariance functions in a wide range of data situations. Our approach can handle dependent processes based on an additive decomposition of the covariance function and is also applicable to processes that are observed on irregular or sparse grids.

In functional data analysis (FDA; see, e.g., Ramsay and Silverman, 2005), covariance functions are at the heart of functional principal component analysis (FPCA), a key tool for dimension reduction based on an eigen analysis of the covariance operator of a random process. FPCA is commonly used to estimate the model parameters in functional predictor and functional response regression models (see Morris, 2015 for an overview). Other examples that are based on covariance functions include functional discriminant analysis (James and Hastie, 2001) and functional canonical correlation analysis (Leurgans et al., 1993). In longitudinal data analysis (LDA), where measurements are frequently recorded at irregularly spaced time points, the correct specification of the covariance benefits the estimation efficiency of the fixed effects and improves the individual predictions (cf. Fan et al., 2007). The covariance is also a crucial ingredient in time series analysis, e.g., in risk models and portfolio allocation (cf. Tai, 2009). The interest commonly lies in a single time series in contrast to FDA (and LDA) where multiple curves are observed, e.g., over time. In principle, our symmetric smoothing approach is also applicable to time series which is, however, not the focus in this paper.

* Correspondence to: Department of Statistics, Faculty of Mathematics, Computer Science and Statistics, LMU Munich, Ludwigstr. 33, 80539 Munich, Germany.
E-mail address: Sonja.Greven@stat.uni-muenchen.de (S. Greven).

https://doi.org/10.1016/j.csda.2017.11.002
0167-9473/© 2017 Elsevier B.V. All rights reserved.

Covariance functions are commonly assumed to be smooth. Thus, when the observed curves are not sufficiently smooth (i.e., observed with error) or not measured on a common dense grid, smoothing becomes necessary at some point during covariance estimation. Directly smoothing the observed curves (see, e.g., Besse and Ramsay, 1986), however, is very difficult or impossible for sparsely observed data which are frequently recorded both in FDA and LDA (Yao et al., 2005). Moreover, pre-smoothing the observed curves removes the measurement error, which is not accounted for in subsequent estimation steps. We pursue an alternative approach and apply bivariate smoothing to the sample covariance of the observed data points.

Most existing work on non-parametric covariance estimation is either restricted to independent functional (or longitudinal) observations and/or only applies to data sampled on a common grid. Furthermore, most bivariate smoothing approaches are not specifically designed for covariances. They do not exploit the symmetry of the estimated surface and thus use redundant information in the available data. To the best of our knowledge, previous approaches have never addressed these issues simultaneously. They can be divided according to three main criteria: (1) the generality of the assumed correlation structure in the data, (2) the generality of possible sampling grids, and (3) the estimation procedure including the selection of the degree of smoothing.

A number of approaches address covariance smoothing in LDA. They are restricted to independent curves but allow for general sampling grids. Smoothing is either accomplished by bivariate kernel smoothing (e.g., Staniswalis and Lee, 1998; Yao et al., 2003, 2005) or by bivariate (penalized) spline smoothing (e.g., Kauermann and Wegener, 2011). The degree of smoothing is either chosen by visual inspection (Staniswalis and Lee, 1998), different leave-one-curve-out cross-validation algorithms (e.g., Yao et al., 2003, 2005) or based on a mixed model representation (e.g., Kauermann and Wegener, 2011). These approaches do not account for the symmetry of the estimated surface. James et al. (2000) directly estimate the smooth eigenfunctions of the covariance function. They estimate a reduced rank mixed effects model via the EM algorithm and use B-spline basis functions to represent the eigenfunctions of the covariance operator. Peng and Paul (2009) estimate the same reduced rank model based on a more efficient Newton–Raphson procedure on the Stiefel manifold. The extension of these reduced rank methods to complex correlation structures is not straightforward. Xiao et al. (2017) recently proposed a bivariate smoother designed for covariance smoothing which can be used for sparsely observed, independent functions. They use bivariate penalized B-splines and enforce a symmetry constraint on the spline coefficients which we take up in our extension to correlated curves. Estimation is done by a three-step procedure which accounts for the covariance of the sample covariance. Their leave-one-curve-out cross-validation procedure for selecting the smoothing parameter is not applicable for correlated functional data, however.

Other covariance smoothing approaches can be applied to correlated functions but are restricted to functions sampled on a common grid and considerably simpler correlation structures than ours. Di et al. (2009) and (Greven et al., 2010) use bivariate penalized splines and select the smoothing parameter using restricted maximum likelihood (REML; Patterson and Thompson, 1971) estimation. Shou et al. (2015) apply a method of moments approach based on symmetric sums represented in a sandwich form. For smoothing, they propose to use an extension of the fast covariance estimation algorithm of Xiao et al. (2016b) to correlated functions. Di et al. (2014) extend the functional random intercept model of Di et al. (2009) to sparsely sampled functional data, but the correlation structure remains less general than ours and an extension is not straightforward. More general correlation structures are allowed in the approach of Cederbaum et al. (2016) that is also suitable for sparsely and irregularly sampled functional data. Their focus lies, however, on a model with crossed functional random effects and estimation is only discussed for this special case. Apart from considering less general correlation structures, all these approaches neither avoid the use of redundant information nor account for the symmetry of the smoothed surface.

We propose a fast symmetric bivariate smoothing approach that applies to data with a broad range of possible correlation structures, much broader than existing methods. Furthermore, our approach is well-suited for (possibly noisy) data sampled on a common, dense grid as well as for irregularly or sparsely sampled data. Strength is borrowed by pooling information across different curves, which is particularly important for curves observed on sparse, unequal grids. The smoothing approach we present is widely applicable: In this paper, we demonstrate how it can be applied to longitudinal data as a special case of independent functional data as well as to correlated functional data with very general and complex correlation structures. For the latter, we extend our bivariate smoothing approach to smoothing additive covariance functions. To the best of our knowledge, all previous proposals in this field have been restricted to estimating much less general dependency structures.

We estimate the covariance functions using a smooth method of moments approach represented as a bivariate additive varying coefficient model. The estimation is based on bivariate penalized splines. We choose the smoothing parameters using REML, which allows the direct extension to additive bivariate smoothing of a superposition of multiple covariance functions. This allows our method to be used for a broad range of complex real-word data settings. It also frees us from having to pre-specify a discrete grid of candidate values for the smoothing parameters that is required for cross-validation based approaches like (Xiao et al., 2017). Smoothing the sample covariance quickly becomes a high-dimensional problem as the number of elements in the sample covariance increases quadratically with the number of grid points. We take advantage of the symmetry of the sample covariance and only estimate the upper triangle of the surface including the diagonal. The estimates are then reflected across the diagonal to obtain the entire estimated covariance, which is continuous but not necessarily smooth across the diagonal. To avoid boundary effects on the diagonal and to ensure identifiability of our models, we enforce smoothness across the diagonal by imposing a symmetry constraint on the spline coefficients, which for the simplest case of independent curves reduces to that of Xiao et al. (2017). We show how the symmetry constraint can be

applied separately to additive covariances and can even be used for any bivariate symmetric smoothing problem beyond covariance functions. Our approach modifies the covariance smoothing approach proposed in Cederbaum et al. (2016) and extends it to more general models. It reduces both the data entering the estimation and the number of spline coefficients that have to be estimated, which leads to considerably faster estimation requiring less memory.

We provide open-source software implementing our approach in the R-package sparseFLMM (R Core Team, 2016; Cederbaum, 2016). The implementation is based on a novel constructor function for the R-package mgcv, which provides a general framework for additive models allowing for a very flexible model specification (Wood, 2011).

We outline the application of our approach to FPCA and demonstrate its practical relevance by an application to sparse longitudinal observations of CD4 cell count trajectories and to densely but irregularly observed acoustic signals from a speech production study. This study requires crossed functional random effects due to repeated measurements for both speakers and target words and thus corresponds to a case of dependent functional data with an additive covariance structure. Both data analyses are fully reproducible as the speech production data are included in the R-package sparseFLMM and the CD4 cell count data are available in the R-package refund (Huang et al., 2016).

We organize our paper as follows: Section 2 first develops our fast symmetric covariance smoothing approach for a simple special case with only one smooth covariance function and additional measurement error. In Section 3, the smoother is extended to complex dependency structures involving the smoothing of multiple additive covariance functions. Section 4 outlines the application of our covariance smoother in FPCA. In Section 5, details on the implementation are given. In Sections 6 and 7, we evaluate our approach in an application to speech production data and in simulations, respectively (a longitudinal application to CD4 cell counts is given in Appendix C). Section 8 closes with a discussion and outlook. Theoretical results and supplementary material are available in the online appendix, where we also provide R-code for our data analyses and simulations based on package sparseFLMM.

2. Fast symmetric covariance smoothing

For simplicity, we first explain our covariance smoothing approach for a simple special case with only one smooth auto-covariance and additional measurement error. This will be extended to additive covariance smoothing in Section 3.

2.1. Model with independent curves

Consider the following model

$$Y_i(t_{ij}) = \mu(t_{ij}, \boldsymbol{x}_i) + E_i(t_{ij}) + \varepsilon_i(t_{ij}), \qquad j = 1, \ldots, D_i, \quad i = 1, \ldots, n, \tag{2.1}$$

where $Y_i(t_{ij})$ is the value of response curve i at the jth observation point $t_{ij} \in \mathcal{T}$, a bounded interval in $\mathbb{R}$. $\mu(t_{ij}, \boldsymbol{x}_i)$ is a global mean function depending on a vector of known covariates $\boldsymbol{x}_i$. $E_i(t_{ij})$ is a smooth curve-specific deviation from the global mean and $\varepsilon_i(t_{ij})$ is additional independent and identically distributed white noise measurement error with constant variance σ^2 that accounts for random uncorrelated variation within curve i. The model can be seen as a function-on-scalar regression model (e.g., Faraway, 1997; Ramsay and Silverman, 2005; Reiss et al., 2010) where all n curves are assumed to be independent and it is a special case of the general functional linear mixed model (FLMM; see Morris, 2015 for a discussion and further references) in Section 3 with only curve-specific smooth residuals. Model (2.1) is often applied to longitudinal data with $\mathcal{T}$ denoting a time interval. Note that all curves may either be observed on a common, fine grid or on curve-specific, possibly sparse, D_i evaluation points $t_{ij}, j = 1, \ldots, D_i, i = 1, \ldots, n$.

In the following, we assume that $E_i(\cdot)$ and $\varepsilon_i(\cdot), i = 1, \ldots, n$, are zero mean, mutually uncorrelated random processes and that $E_i(\cdot)$ is square-integrable. We denote the auto-covariance function of $E_i(\cdot)$ by $K^E(t, t') = \text{Cov}\left[E_i(t), E_i(t')\right], t, t' \in \mathcal{T}$. We further assume that the mean and the auto-covariance are smooth in t and in arguments t, t', respectively.

2.2. Estimation in the independent case

We apply the following smooth method of moments approach to estimate the auto-covariance $K^E(t, t')$. It modifies the approach presented in Cederbaum et al. (2016) by accounting for the symmetry of covariances, which leads to a considerable reduction of computation times. In this simple case of independent functional responses, our smoother is closely related to that of Xiao et al. (2017), who approach the problem from a slightly different perspective. While their focus is mainly on the symmetry constraint for the spline coefficients and on the development of a fast smoothing parameter estimation, our aim is to avoid redundant information in symmetric smoothing which in this simple case leads us to the same symmetry constraint. Their approach, however, is not directly extendable to correlated data as will be discussed at the end of this section.

We focus in the following on the centered functional responses $\widetilde{Y}_i(t_{ij}) = Y_i(t_{ij}) - \mu(t_{ij}, \boldsymbol{x}_i)$ with expectation zero and denote their realizations by $\widetilde{y}_{it_{ij}}$. We exploit the fact that the expectation of the centered cross products corresponds to the covariance of the functional response which is given as

$$\begin{aligned} &\mathbb{E}\left[\widetilde{Y}_i(t_{ij})\widetilde{Y}_i(t_{ij'})\right] = \text{Cov}\left[Y_i(t_{ij}), Y_i(t_{ij'})\right] = K^E(t_{ij}, t_{ij'}) + \sigma^2\delta_{jj'}, \\ &j, j' = 1, \ldots, D_i, \quad i = 1, \ldots, n, \end{aligned} \tag{2.2}$$

with $\delta_{jj'}$ equal to one if $j = j'$ and zero otherwise. Eq. (2.2) can be seen as a special case of a bivariate additive varying coefficient model for the empirical covariances $\widetilde{y}_{it_{ij}}\widetilde{y}_{it_{ij'}}$, in which the auto-covariance and the error variance are the unknown components. We estimate the smooth auto-covariance function $K^E(t, t')$ and the error variance σ^2 simultaneously using a bivariate spline representation for $K^E(t, t')$ under working assumptions of independence and homoscedasticity.

For this, let $\boldsymbol{C}$ denote the $\mathcal{C}\times 1$ stacked vector of all cross products, with $\mathcal{C} = \sum_{i=1}^{n} D_i^2$. Then, Model (2.2) can be represented as

$$\mathbb{E}(\boldsymbol{C}) = \left[\boldsymbol{M}^E | \boldsymbol{\delta}^{\varepsilon}\right] \left(\boldsymbol{\theta}^{E\top}, \sigma^2\right)^{\top} =: \boldsymbol{M}\boldsymbol{\alpha}, \tag{2.3}$$

where $\boldsymbol{M}^E$ denotes the $\mathcal{C} \times \left(F^E\right)^2$ bivariate spline design matrix, containing the evaluations of any bivariate spline basis with $\left(F^E\right)^2$ basis functions that are symmetric across the diagonal, i.e., across $t_{ij} = t_{ij'}$. We use bivariate tensor product B-splines, but other bases are possible. See Section 5 and Appendix B for details. $\boldsymbol{\delta}^{\varepsilon}$ is an indicator vector of length $\mathcal{C}$ whose elements take values $\delta_{jj'}$. $\boldsymbol{\theta}^E$ is a spline coefficient vector of length $\left(F^E\right)^2$. To avoid over-fitting, we use an isotropic quadratic smoothness penalty of the form

$$\text{pen}(\lambda) = \lambda \boldsymbol{\theta}^{E\top} \boldsymbol{S}^E \boldsymbol{\theta}^E, \tag{2.4}$$

where λ denotes the smoothing parameter that controls the bias–variance tradeoff and $\boldsymbol{S}^E$ is a suitable penalty matrix of dimension $\left(F^E\right)^2 \times \left(F^E\right)^2$, see Section 5 for a discussion.

The development above uses all $\mathcal{C}$ cross products for the estimation of the covariance as is commonly done in published smooth method of moments approaches for covariance estimation [Yao et al., 2003; Yao et al., 2005; for correlated curves, e.g., Staniswalis and Lee, 1998; Di et al., 2009; Greven et al., 2010]. As it is quadratic in the number of function evaluations, $\mathcal{C}$ quickly becomes extremely large in practice and often poses significant computational challenges. Since the cross products $\widetilde{y}_{it_{ij}}\widetilde{y}_{it_{ij'}}$ and $\widetilde{y}_{it_{ij'}}\widetilde{y}_{it_{ij}}$ are identical, we, like (Xiao et al., 2017), avoid the use of this redundant information and only estimate the upper triangle of the auto-covariance surface including the diagonal. Using the identical cross products only once also decreases the dependency among the cross products in $\boldsymbol{C}$ and thus reduces the strongest violations of the implicit working assumption of independent observations in our additive model (2.3). A detailed discussion of working assumptions and possible strategies for handling violations is given in Section 3.4.

We assume in the following that $\boldsymbol{C}$ is sorted such that it can be partitioned as $\boldsymbol{C} = \left(\boldsymbol{C}_{t<t'}^{\top}, \boldsymbol{C}_{t=t'}^{\top}, \boldsymbol{C}_{t>t'}^{\top}\right)^{\top}$, where $\boldsymbol{C}_{t<t'}$, $\boldsymbol{C}_{t=t'}$, $\boldsymbol{C}_{t>t'}$ comprise all cross products $\widetilde{Y}_{it_{ij}}\widetilde{Y}_{it_{ij'}}$, $i = 1, \ldots, n$, with $t_{ij} < t_{ij'}$, $t_{ij} = t_{ij'}$, and $t_{ij} > t_{ij'}$, respectively. Then, with suitable sorting within the three partitions of $\boldsymbol{C}$, the symmetry of the cross products implies that $\boldsymbol{C}_{t<t'} = \boldsymbol{C}_{t>t'}$. In order to speed up estimation, we only use $\boldsymbol{C}^{\Delta} := \left(\boldsymbol{C}_{t<t'}^{\top}, \boldsymbol{C}_{t=t'}^{\top}\right)^{\top}$ for the estimation in the bivariate model (2.3). The total number of cross products thus amounts to $\mathcal{C}^{\Delta} := \sum_{i=1}^{n} D_i(D_i+1)/2$. The design matrix of the bivariate additive model can accordingly be partitioned as $\boldsymbol{M} = \left(\boldsymbol{M}_{t<t'}^{\top}, \boldsymbol{M}_{t=t'}^{\top}, \boldsymbol{M}_{t>t'}^{\top}\right)^{\top}$. Let $\boldsymbol{M}^{E\Delta}$ and $\boldsymbol{\delta}^{\epsilon\Delta}$ denote the submatrix and subvector corresponding to $\boldsymbol{C}^{\Delta}$. Then, the bivariate additive model (2.3) reduces to $\mathbb{E}\left(\boldsymbol{C}^{\Delta}\right) = \left[\boldsymbol{M}^{E\Delta} | \boldsymbol{\delta}^{\varepsilon\Delta}\right] \left(\boldsymbol{\theta}^{E\top}, \sigma^2\right)^{\top}$. Reflecting the upper triangle across the diagonal ensures that the obtained surface estimate is symmetric and continuous in both directions t and t', but it does not guarantee smoothness across the diagonal without additional constraint. Wiggly estimates in the area of the diagonal can occur as coefficients for basis functions near the diagonal have few observations available for estimation. This is because much of the support of the basis functions is on the other side of the diagonal, i.e., in the lower triangle that does not contain any data. An expected consequence is that the smooth auto-covariance surface and the error variance on the diagonal can be less well separated, which we indeed observe in our application in Section 6 and in our simulations in Section 7.

We enforce smoothness across the diagonal in order to avoid such boundary effects. The assumption of smoothness across the diagonal guarantees identifiability of our model as the smooth auto-covariance $K^E(t_{ij}, t_{ij'})$ and the variance of the additional unsmooth error term $\varepsilon_i(t_{ij})$ can be separated. A symmetric surface implies a symmetric spline coefficient matrix $\boldsymbol{\Theta}^E = \left[\theta^E_{bb'}\right]_{b,b'=1,\ldots,F^E} = \boldsymbol{\Theta}^{E\top}$, where $\boldsymbol{\theta}^E = \left(\boldsymbol{\theta}^{E\top}_{b<b'}, \boldsymbol{\theta}^{E\top}_{b=b'}, \boldsymbol{\theta}^{E\top}_{b>b'}\right)^{\top}$ contains first the entries of $\boldsymbol{\Theta}^E$ below the diagonal ($\theta^E_{bb'}$, $b < b'$), then the diagonal entries ($\theta^E_{bb'}$, $b = b'$) and lastly the entries above the diagonal ($\theta^E_{bb'}$, $b > b'$). We impose a symmetry constraint on $\boldsymbol{\Theta}^E$. Thus, our approach differs in two crucial points from the naive covariance estimation in (2.3) and from most previous covariance smoothing approaches. First, we reduce the number of cross products that enter the estimation and second, imposing the symmetry constraint almost halves the number of spline coefficients that have to be estimated. Both aspects greatly speed up the computation as we show in Sections 6 and 7.

With suitable sorting within the partitions $\boldsymbol{\theta}^E_{b<b'}$ and $\boldsymbol{\theta}^E_{b>b'}$, the above symmetry constraint on the coefficient matrix corresponds to the following symmetry constraint on the coefficient vector

$$\theta^E_{bb'} = \theta^E_{b'b}, \quad b, b' = 1, \ldots, F^E \qquad \Leftrightarrow \qquad \boldsymbol{\theta}^E_{b<b'} = \boldsymbol{\theta}^E_{b>b'}, \tag{2.5}$$

which corresponds to the constraint used in Xiao et al. (2017). This allows us to consider the reduced coefficient vector $\boldsymbol{\theta}^{Er} = \left(\boldsymbol{\theta}^{E\top}_{b<b'}, \boldsymbol{\theta}^{E\top}_{b=b'}\right)^{\top}$ of length $F^E\left(F^E+1\right)/2$. For the implementation of the above symmetry constraint, we use that under the constraint we have $\boldsymbol{M}^{E\Delta}\boldsymbol{\theta}^E = \left(\boldsymbol{M}^{E\Delta}_{b<b'} + \boldsymbol{M}^{E\Delta}_{b>b'}\right)\boldsymbol{\theta}^E_{b<b'} + \boldsymbol{M}^{E\Delta}_{b=b'}\boldsymbol{\theta}^E_{b=b'}$, with $\boldsymbol{M}^{E\Delta}_{b<b'}$, $\boldsymbol{M}^{E\Delta}_{b=b'}$, and $\boldsymbol{M}^{E\Delta}_{b>b'}$ containing the

respective columns of $\boldsymbol{M}^{E\Delta}$. Thus, the constraint is equivalent to adding up columns $\boldsymbol{M}^{E\Delta}_{b<b'}$ and $\boldsymbol{M}^{E\Delta}_{b>b'}$ of the design matrix. This can be achieved by right multiplication of $\boldsymbol{M}^{E\Delta}$ with the $\left(F^E\right)^2 \times F^E\left(F^E+1\right)/2$ constraint matrix

$$\boldsymbol{W}^E = \begin{bmatrix} \boldsymbol{I}_{\frac{F^E(F^E-1)}{2}} & \boldsymbol{0}_{\frac{F^E(F^E-1)}{2}\times F^E} \\ \boldsymbol{0}_{F^E\times\frac{F^E(F^E-1)}{2}} & \boldsymbol{I}_{F^E} \\ \boldsymbol{I}_{\frac{F^E(F^E-1)}{2}} & \boldsymbol{0}_{\frac{F^E(F^E-1)}{2}\times F^E} \end{bmatrix},$$

where $\boldsymbol{I}_x$ is an identity matrix of dimension x and $\boldsymbol{0}_{x\times y}$ is a null matrix of dimension $x \times y$. We denote the reduced design matrix by $\boldsymbol{M}^{E\Delta r} := \boldsymbol{M}^{E\Delta}\boldsymbol{W}^E$. Under the symmetry constraint, $\boldsymbol{M}^{E\Delta r}\boldsymbol{\theta}^{Er} = \boldsymbol{M}^{E\Delta}\boldsymbol{\theta}^E$. The penalty matrix in (2.4) also needs to be adjusted to the reduced coefficient vector $\boldsymbol{\theta}^{Er}$ as $\boldsymbol{S}^{Er} := \boldsymbol{W}^{E\top}\boldsymbol{S}^E\boldsymbol{W}^E$, corresponding to $\boldsymbol{\theta}^{Er\top}\boldsymbol{S}^{Er}\boldsymbol{\theta}^{Er} = \boldsymbol{\theta}^{E\top}\boldsymbol{S}^E\boldsymbol{\theta}^E$.

Altogether, the bivariate additive model for the reduced response vector with symmetry constraint (2.5) is given by

$$\mathbb{E}\left(\boldsymbol{C}^{\Delta}\right) = \left[\boldsymbol{M}^{E\Delta r}|\boldsymbol{\delta}^{\varepsilon\Delta}\right]\left(\boldsymbol{\theta}^{Er\top}, \sigma^2\right)^{\top} =: \boldsymbol{M}^{\Delta r}\boldsymbol{\alpha}^r. \tag{2.6}$$

Note that in Model (2.6), each product $\widetilde{y}_{it_{ij}}\widetilde{y}_{it_{ij'}}$, $t_{ij} \leq t_{ij'}$, enters the estimation with the same weight. This is not the case when all products $\widetilde{y}_{it_{ij}}\widetilde{y}_{it_{ij'}}$, $j, j' = 1, \ldots, D_i$, are used as in Model (2.3), where all products appear twice except for those on the diagonal ($t_{ij} = t_{ij'}$). Our implementation in R-package sparseFLMM allows to estimate Model (2.6) with the same weights as in Model (2.3) by putting a weight of 0.5 on the products on the diagonal. There is room for debate on whether it is desirable to down-weigh the data on the diagonal compared to the rest. In any case, our simulations in Section 7 show that the difference is not very large.

In contrast to Xiao et al. (2017), who derive a leave-one-curve-out generalized cross-validation (GCV) algorithm to choose the smoothing parameter for independent curves, we choose the smoothing parameter as variance component ratio using REML. REML has been shown to be more stable than GCV and to result in lower mean squared errors (Reiss and Ogden, 2009; Wood, 2011). Moreover, Krivobokova and Kauermann (2007) have shown that REML based smoothing parameter selection is less sensitive towards misspecifications of the correlation structure than prediction error methods. Even more importantly, using REML allows us to directly extend our symmetric smoothing approach to additive smoothing needed for functional data with complex dependency structures as will be shown in Section 3. In more general designs, where the responses cannot be decomposed into independent subvectors, it is not clear how to perform smoothing parameter selection based on GCV and optimizing multiple smoothing parameters would require a computationally costly multi-dimensional grid search.

Details on the implementation in the R-package sparseFLMM are given in Section 5 and in Appendix B.

3. Fast symmetric additive covariance smoothing

Simultaneous REML estimation of multiple smoothing parameters allows the direct extension of our approach to more general models with complex correlation structures, for which we derive appropriate symmetry constraint matrices.

3.1. General functional linear mixed model

The general FLMM (see, e.g., Morris, 2015) can be seen as the functional analogue of the linear mixed model (LMM; see, e.g., Pinheiro and Bates, 2000), which is often applied to scalar correlated data. The random effects in the linear mixed model are replaced by functional random effects in order to account for the functional nature of the response. A functional random intercept (fRI) for a subject, for example, is a subject-specific deviation from the mean in the form of a function. The FLMM is given by

$$\begin{aligned} Y_i(t_{ij}) &= \mu(t_{ij}, \boldsymbol{x}_i) + \boldsymbol{z}_i^{\top}\boldsymbol{U}(t_{ij}) + E_i(t_{ij}) + \varepsilon_i(t_{ij}), \\ &j = 1, \ldots, D_i, \quad i = 1, \ldots, n, \end{aligned} \tag{3.1}$$

where $Y_i(t_{ij})$ denotes the response of curve i at observation point t_{ij}, which can be additively decomposed as in Model (2.1). Model (3.1), however, additionally accounts for correlation between (groups of) curves by the vector-valued random process $\boldsymbol{U}(t_{ij})$ which is multiplied by $\boldsymbol{z}_i$, a known covariate vector of length q. Examples for $\boldsymbol{z}_i^{\top}\boldsymbol{U}(t_{ij})$ yielding FLMMs with, e.g., crossed and hierarchical functional random effects are given in Section 3.3 and in Appendix B.

We assume that $\boldsymbol{U}(\cdot)$, $E_i(\cdot)$, and $\varepsilon_i(\cdot)$ are zero mean, mutually uncorrelated random processes and that $\boldsymbol{U}(\cdot)$ and $E_i(\cdot)$ are square-integrable. As for Model (2.1), we denote the auto-covariance of $E_i(\cdot)$ by $K^E(t, t') = \mathrm{Cov}\left[E_i(t), E_i(t')\right]$, $t, t' \in \mathcal{T}$. The $q \times q$ matrix-valued auto-covariance of $\boldsymbol{U}(\cdot)$ is denoted by $\boldsymbol{K}^U(t, t') = \mathrm{Cov}\left[\boldsymbol{U}(t), \boldsymbol{U}(t')\right]$. The covariances are assumed to be smooth (for each component in the case of $\boldsymbol{U}(t)$).

Let G denote the number of grouping variables. Then, $\boldsymbol{U}(t_{ij})$ can be divided into G independent blocks $\boldsymbol{U}_g(t_{ij})$, $g = 1, \ldots, G$, which again contain blocks of L^{U_g} independent copies $\boldsymbol{U}_{gl}(t_{ij})$, $l = 1, \ldots, L^{U_g}$, where L^{U_g} is the number of levels of the gth grouping variable. $\boldsymbol{U}_{gl}(t_{ij}) = \left[U_{gl1}(t_{ij}), \ldots, U_{gl\rho^{U_g}}(t_{ij})\right]^{\top}$, in turn, is a vector-valued random process of ρ^{U_g} components for each level of this grouping variable, for example $\rho^{U_g} = 2$ if the gth grouping variable is associated with a fRI and a functional

random slope. The total number of entries in $\boldsymbol{U}(t_{ij})$ is given by $q = \sum_{g=1}^{G} L^{U_g} \rho^{U_g}$. The $\rho^{U_g} \times \rho^{U_g}$ matrix-valued covariance of $\boldsymbol{U}_{gl}(\cdot)$, $\boldsymbol{K}^{U_g}(t,t') = \left[K^{U_g}_{ss'}(t,t')\right]_{s,s'=1,\ldots,\rho^{U_g}} = \mathrm{Cov}\left[\boldsymbol{U}_{gl}(t), \boldsymbol{U}_{gl}(t')\right]$, with $K^{U_g}_{ss'}(t,t') = K^{U_g}_{s's}(t',t)$, is the same for all levels, $l = 1,\ldots,L^{U_g}$, of the gth grouping variable. We can thus write the block-diagonal auto-covariance of $\boldsymbol{U}(\cdot)$ as

$$\boldsymbol{K}^{U}(t,t') = \mathrm{diag}\left(\underbrace{\boldsymbol{K}^{U_1}(t,t'),\ldots,\boldsymbol{K}^{U_1}(t,t')}_{L^{U_1}\text{ times}},\ldots,\underbrace{\boldsymbol{K}^{U_G}(t,t'),\ldots,\boldsymbol{K}^{U_G}(t,t')}_{L^{U_G}\text{ times}}\right).$$

3.2. Estimation in the general functional linear mixed model

Our fast symmetric covariance smoothing approach can be extended to the general model (3.1) by generalizing it to a matrix of covariances as described above and applying it to each additive component separately.

In analogy to Model (2.1), we base the covariance smoothing on the following decomposition of the expectation of the cross products of the centered functional responses

$$\begin{aligned}\mathbb{E}\left[\widetilde{Y}_i(t_{ij})\widetilde{Y}_{i'}(t_{i'j'})\right] &= \mathrm{Cov}\left[Y_i(t_{ij}), Y_{i'}(t_{i'j'})\right] \\ &= \boldsymbol{z}_i^{\top}\boldsymbol{K}^{U}(t_{ij},t_{i'j'})\boldsymbol{z}_{i'} + \left[K^{E}(t_{ij},t_{i'j'}) + \sigma^2\delta_{jj'}\right]\delta_{ii'},\end{aligned} \tag{3.2}$$

where, in contrast to Model (2.1), products are now also computed across different curves i, i'.

Let $\ell_g(i)$ denote the level of grouping variable g for observation i. Similar to $\boldsymbol{U}(\cdot)$, the covariate vector $\boldsymbol{z}_i$ can be divided into G blocks $\boldsymbol{z}_i^{\top} = \left(\boldsymbol{z}_i^{U_1\top},\ldots,\boldsymbol{z}_i^{U_G\top}\right)$, where the blocks $\boldsymbol{z}_i^{U_g}$, $g = 1,\ldots,G$, can again be written as $\boldsymbol{z}_i^{U_g\top} = \left(\boldsymbol{z}_{i1}^{U_g\top},\ldots,\boldsymbol{z}_{iL^{U_g}}^{U_g\top}\right)$ with $\boldsymbol{z}_{il}^{U_g\top} = \left(z_{il1}^{U_g},\ldots,z_{il\rho^{U_g}}^{U_g}\right)$, $l = 1,\ldots,L^{U_g}$. The scalars $z_{ils}^{U_g}$ take the value of the respective covariate $\omega_{is}^{U_g}$ times an indicator $\delta_{\ell_g(i)l}$, specifying whether observation i belongs to level l of grouping variable g. Based on this partition, the expectation in Model (3.2) can be rewritten as

$$\begin{aligned}\mathbb{E}\left[\widetilde{Y}_i(t_{ij})\widetilde{Y}_{i'}(t_{i'j'})\right] &= \sum_{g=1}^{G}\sum_{l=1}^{L^{U_g}}\sum_{s=1}^{\rho^{U_g}}\sum_{s'=1}^{\rho^{U_g}} z_{ils}^{U_g} z_{i'ls'}^{U_g} K^{U_g}_{ss'}(t_{ij},t_{i'j'}) \\ &\quad + \left[K^{E}(t_{ij},t_{i'j'}) + \sigma^2\delta_{jj'}\right]\delta_{ii'}.\end{aligned} \tag{3.3}$$

For example in the case of an FLMM with only one fRI ($G = 1$, $\rho^{U_1} = 1$), the products $z_{ils}^{U_g} z_{i'ls'}^{U_g}$ yield indicators for whether the two observations in a cross product belong to the same level of the grouping variable.

We exploit the symmetry of covariances $\boldsymbol{K}^{U_g}(t_{ij},t_{i'j'}) = \boldsymbol{K}^{U_g}(t_{i'j'},t_{ij})^{\top}$, $g = 1,\ldots,G$, and of $K^{E}(t_{ij},t_{i'j'}) = K^{E}(t_{i'j'},t_{ij})$ and only use the products $\widetilde{Y}_{it_{ij}}\widetilde{Y}_{i't_{i'j'}}$ with $t_{ij} \le t_{i'j'}$ if $i \le i'$ and $t_{ij} < t_{i'j'}$, otherwise, suitably sorted in the long vector $\boldsymbol{C}^{\Delta}$. Let $\cdot$ denote the Hadamard (pointwise) product. As in the case with independent curves, Model (3.3) can be represented as bivariate additive varying coefficient model, here of the form

$$\begin{aligned}&\mathbb{E}\left(\boldsymbol{C}^{\Delta}\right) = \\ &\qquad\left[\boldsymbol{M}^{U_1\Delta}|\ldots|\boldsymbol{M}^{U_G\Delta}|\boldsymbol{M}^{E\Delta}|\boldsymbol{\delta}^{\varepsilon\Delta}\right]\left(\boldsymbol{\theta}^{U_1\top},\ldots,\boldsymbol{\theta}^{U_G\top},\boldsymbol{\theta}^{E\top},\sigma^2\right)^{\top},\end{aligned} \tag{3.4}$$

where $\boldsymbol{M}^{U_g\Delta}$, $g = 1,\ldots,G$, contain the column-wise concatenated submatrices $\boldsymbol{M}^{U_g\Delta}_{ss'}$, corresponding to the covariances $K^{U_g}_{ss'}(t,t')$, $s,s' = 1,\ldots,\rho^{U_g}$. The submatrices $\boldsymbol{M}^{U_g\Delta}_{ss'}$ are given by $\boldsymbol{Q}^{U_g\Delta}_{ss'} \cdot \boldsymbol{B}^{U_g\Delta}_{ss'}$, where $\boldsymbol{Q}^{U_g\Delta}_{ss'}$ contain suitably sorted and repeated entries $\delta_{\ell_g(i)\ell_g(i')} \cdot \omega^{U_g}_{is}\omega^{U_g}_{i's'}$ and $\boldsymbol{B}^{U_g\Delta}_{ss'}$ denote the bivariate spline design matrices. $\boldsymbol{M}^{E\Delta}$ is analogously given by $\boldsymbol{Q}^{E\Delta} \cdot \boldsymbol{B}^{E\Delta}$, with bivariate spline design matrix $\boldsymbol{B}^{E\Delta}$ and $\boldsymbol{Q}^{E\Delta}$ a new indicator matrix, which reduces to an all-ones matrix in the model with independent curves, for which thus $\boldsymbol{M}^{E\Delta} = \boldsymbol{B}^{E\Delta}$. The concrete form of $\boldsymbol{Q}^{U_g\Delta}_{ss'}$ and $\boldsymbol{Q}^{E\Delta}$, as well as the bivariate spline design matrices for tensor product B-splines are provided in Appendix B.

We assume that for each $g = 1,\ldots,G$, the coefficient vector $\boldsymbol{\theta}^{U_g}$ is sorted with the order corresponding to the columns of

$$\boldsymbol{M}^{U_g\Delta} = \left[\boldsymbol{M}^{U_g\Delta}_{11}|\ldots|\boldsymbol{M}^{U_g\Delta}_{1\rho^{U_g}}|\ldots|\boldsymbol{M}^{U_g\Delta}_{\rho^{U_g}1}|\ldots|\boldsymbol{M}^{U_g\Delta}_{\rho^{U_g}\rho^{U_g}}\right].$$

Moreover, with suitable sorting, each submatrix $\boldsymbol{M}^{U_g\Delta}_{ss'}$ can be partitioned as in the case of independent curves, $\boldsymbol{M}^{U_g\Delta}_{ss'} = \left[\boldsymbol{M}^{U_g\Delta}_{ss',b<b'}|\boldsymbol{M}^{U_g\Delta}_{ss',b=b'}|\boldsymbol{M}^{U_g\Delta}_{ss',b>b'}\right]$. Let $\boldsymbol{\Theta}^{U_g}_{ss'}$ denote the coefficient matrices, where the correspondingly sorted coefficient vectors $\boldsymbol{\theta}^{U_g}_{ss'} = \left(\boldsymbol{\theta}^{U_g\top}_{ss',b<b'},\boldsymbol{\theta}^{U_g\top}_{ss',b=b'},\boldsymbol{\theta}^{U_g\top}_{ss',b>b'}\right)^{\top}$ contain first the entries of the spline coefficient matrix $\boldsymbol{\Theta}^{U_g}_{ss'}$ below the diagonal, then the diagonal entries and lastly the entries above the diagonal. Assume further that within the three blocks of $\boldsymbol{\theta}^{U_g}_{ss'}$, the entries $\theta_{ss',bb'}$ are sorted correspondingly for all $s,s' = 1,\ldots,\rho^{U_g}$.

As a modular component, the symmetry constraint

$$\boldsymbol{\Theta}_{ss'}^{U_g} = \boldsymbol{\Theta}_{s's}^{U_g\top}, \qquad s, s' = 1, \ldots, \rho^{U_g}, \tag{3.5}$$

can be applied to each, $g = 1, \ldots, G$, due to the symmetry of covariances $\boldsymbol{K}^{U_g}(t, t')$, yielding the reduced coefficient vectors $\boldsymbol{\theta}_{ss'}^{U_g r}$, $s \leq s'$, and thus the reduced long coefficient vector $\boldsymbol{\theta}^{U_g r}$. As in the case of independent curves, the constraint (3.5) is equivalent to adding up the respective columns of the large design matrix $\boldsymbol{M}^{U_g \Delta}$. This can be achieved by right-multiplication of $\boldsymbol{M}^{U_g \Delta}$ with a suitable constraint matrix $\boldsymbol{W}^{U_g}$, yielding the reduced design matrix $\boldsymbol{M}^{U_g \Delta r} = \left[\boldsymbol{M}_{11}^{U_g \Delta r} | \ldots | \boldsymbol{M}_{1\rho^{U_g}}^{U_g \Delta r} | \boldsymbol{M}_{22}^{U_g \Delta r} | \ldots | \boldsymbol{M}_{\rho^{U_g}-1\rho^{U_g}-1}^{U_g \Delta r} | \boldsymbol{M}_{\rho^{U_g}-1\rho^{U_g}}^{U_g \Delta r} | \boldsymbol{M}_{\rho^{U_g}\rho^{U_g}}^{U_g \Delta r}\right]$.

Each $\boldsymbol{M}_{ss'}^{U_g \Delta r}$, $s \leq s'$, consists of column-wise concatenated matrices

$$\boldsymbol{M}_{ss'}^{U_g \Delta r} = \left[\boldsymbol{M}_{ss',b<b'}^{U_g \Delta} + \boldsymbol{M}_{s's,b>b'}^{U_g \Delta} | \boldsymbol{M}_{ss',b=b'}^{U_g \Delta} + \boldsymbol{M}_{s's,b=b'}^{U_g \Delta} \delta_{s<s'}\right],$$
$$s \leq s' = 1, \ldots, \rho^{U_g}.$$

The constraint matrix $\boldsymbol{W}^{U_g}$ consists of $\left(\rho^{U_g}\right)^2 \times \frac{\left(\rho^{U_g}\right)^2+1}{2}$ blocks, most of which are zero. The block-rows of $\boldsymbol{W}^{U_g}$ correspond to the constraint on the spline coefficients of the covariances $K_{ss'}^{U_g}(t, t')$, $s, s' = 1, \ldots, \rho^{U_g}$, sorted as in $\boldsymbol{M}^{U_g \Delta}$. The columns are sorted correspondingly to the reduced matrix $\boldsymbol{M}^{U_g \Delta r}$. For the auto-covariances ($s = s'$), the blocks are of the same form as the constraint matrix $\boldsymbol{W}^{E}$ for independent curves. For the cross-covariances ($s < s'$), the blocks either correspond to diagonal block matrices or to anti-diagonal block matrices, depending on whether the respective rows correspond to $s < s'$ or $s > s'$, respectively. The specific form of $\boldsymbol{W}^{U_g}$ and examples for $\rho^{U_g} = 2, 3$ are given in Appendix B.

A quadratic smoothness penalty associated with each smooth term controls the bias–variance tradeoff. Each penalty matrix $\boldsymbol{S}^{U_g}$, consisting of blocks for each $K_{ss'}^{U_g}(t, t')$, is accordingly reduced by left-and right-multiplication with the constraint matrix $\boldsymbol{W}^{U_g}$. Smoothing the components in the upper triangle of $\boldsymbol{K}^{U_g}(t, t')$ separately allows to define different penalties for the auto-covariances and the cross-covariances, respectively. In particular, it is possible to apply anisotropic penalties for the cross-covariances.

Reflecting the estimated triangular covariance surfaces across the diagonal yields estimates for the whole covariance surfaces $\boldsymbol{K}^{U_g}(t, t')$ and $K^{E}(t, t')$, with smoothness assured also across the diagonal. Note that reducing computation time becomes even more important when multiple covariances are smoothed simultaneously rather than a single one as in the case of independent functional data in Section 2.

3.3. Functional linear mixed model with crossed random intercepts

Motivated by our application to the phonetics data in Section 6.2, we now illustrate the specification of the FLMM for the special case of an FLMM with crossed fRIs accounting for the repeated measurements on two grouping variables (e.g., speakers and target words) in a crossed design. In this model, we have $G = 2$ grouping variables with $\rho^{U_1} = \rho^{U_2} = 1$ associated random effects for each grouping variable, i.e., one fRI each. L^{U_1}, L^{U_2} are the numbers of levels of the first (e.g., speakers) and second (e.g., target words) grouping variable, respectively. The covariate vector $\boldsymbol{z}_i$ only consists of indicators taking value one or zero to code group membership for the two grouping variables. The explicit specification of the covariate vector for crossed and hierarchical functional random effects is given in Appendix B. For better readability, we rename in the following the components of the vector-valued random process as $B := U_1$ and $C := U_2$. The total number of components in $\boldsymbol{U}(t_{ij})$ is $q = L^B + L^C$. Note that this model corresponds to the model in Cederbaum et al. (2016).

For this model, Eq. (3.3) can be simplified as

$$\begin{aligned}\mathbb{E}\left[\widetilde{Y}_i(t_{ij})\widetilde{Y}_{i'}(t_{i'j'})\right] &= K^B(t_{ij}, t_{i'j'})\delta_{\ell_1(i)\ell_1(i')} \\ &+ K^C(t_{ij}, t_{i'j'})\delta_{\ell_2(i)\ell_2(i')} \\ &+ \left[K^E(t_{ij}, t_{i'j'}) + \sigma^2\delta_{jj'}\right]\delta_{ii'},\end{aligned} \tag{3.6}$$

where $\delta_{\ell_1(i)\ell_1(i')}$ and $\delta_{\ell_2(i)\ell_2(i')}$ take value one when the two curves i and i' belong to the same level of the respective grouping variable and zero otherwise. As can be seen in Eq. (3.6), the products for which neither $\delta_{\ell_1(i)\ell_1(i')}$ nor $\delta_{\ell_2(i)\ell_2(i')}$ equals one do not have to be considered due to expectation zero. Eq. (3.4) then reduces to

$$\mathbb{E}\left(\boldsymbol{C}^{\Delta}\right) = \left[\boldsymbol{M}^{B\Delta} | \boldsymbol{M}^{C\Delta} | \boldsymbol{M}^{E\Delta} | \boldsymbol{\delta}^{\varepsilon\Delta}\right]\left(\boldsymbol{\theta}^{B\top}, \boldsymbol{\theta}^{C\top}, \boldsymbol{\theta}^{E\top}, \sigma^2\right)^{\top} \tag{3.7}$$

and the symmetry constraint can be applied to each $\boldsymbol{\theta}^B$, $\boldsymbol{\theta}^C$, and $\boldsymbol{\theta}^E$.

3.4. Covariance of cross products

We estimate the auto-covariances as unknown, smooth functions in a bivariate additive varying coefficient model using a quadratic loss function. Since this is equivalent to a penalized likelihood criterion for Gaussian data, we implicitly assume

independence of the cross products with homoscedastic Gaussian measurement error. As already mentioned in Section 2.2 and shortly discussed in Cederbaum et al. (2016), these are working assumptions which do not hold as the products frequently involve two points on the same curve or on correlated curves. Nevertheless, these implicit assumptions are made by many existing works (e.g., Yao et al., 2005; Di et al., 2009; Greven et al., 2010). We now briefly discuss how the covariance of the cross products can be accounted for.

For the model with independent curves (2.1), (Xiao et al., 2017) derive an expression for the covariance of the cross products in terms of $K^E(t, t')$ and the error variance σ^2 under the assumption of Gaussian responses. They apply a three-step algorithm in which they first estimate $K^E(t, t')$ and σ^2 under the working assumptions. Second, they estimate the covariance of the cross products by plugging in the estimates for $K^E(t, t')$ and σ^2. In the third step, they re-estimate $K^E(t, t')$ and σ^2 using the estimated covariance of the cross products as a working covariance.

We derive an expression for the covariance of the cross products for the general model (3.1) based on results from (Isserlis, 1918) on fourth moment rules for multivariate Gaussian random variables. The covariance of the cross products can be written as

$$\begin{aligned}
&\operatorname{Cov}\left[\widetilde{Y}_i(t_{ij})\widetilde{Y}_{i'}(t_{i'j'}), \widetilde{Y}_m(t_{mo})\widetilde{Y}_{m'}(t_{m'o'})\right] \\
&= \left\{\boldsymbol{z}_i^\top \boldsymbol{K}^U(t_{ij}, t_{mo})\boldsymbol{z}_m + \left[K^E(t_{ij}, t_{mo}) + \sigma^2\delta_{jo}\right]\delta_{im}\right\} \\
&\quad \cdot \left\{\boldsymbol{z}_{i'}^\top \boldsymbol{K}^U(t_{i'j'}, t_{m'o'})\boldsymbol{z}_{m'} + \left[K^E(t_{i'j'}, t_{m'o'}) + \sigma^2\delta_{j'o'}\right]\delta_{i'm'}\right\} \\
&+ \left\{\boldsymbol{z}_i^\top \boldsymbol{K}^U(t_{ij}, t_{m'o'})\boldsymbol{z}_{m'} + \left[K^E(t_{ij}, t_{m'o'}) + \sigma^2\delta_{jo'}\right]\delta_{im'}\right\} \\
&\quad \cdot \left\{\boldsymbol{z}_{i'}^\top \boldsymbol{K}^U(t_{i'j'}, t_{mo})\boldsymbol{z}_m + \left[K^E(t_{i'j'}, t_{mo}) + \sigma^2\delta_{j'o}\right]\delta_{i'm}\right\}.
\end{aligned} \tag{3.8}$$

The derivation and simplifications for the model with crossed fRIs are given in Appendix A. The covariance (3.8) is a function of the unknown covariances $\boldsymbol{K}^U(t, t'), K^E(t, t')$ and σ^2, giving rise to the need for an iterative procedure as proposed in Xiao et al. (2017) for the simpler model (2.1). Our implementation with R-package mgcv allows to directly include the variance of the cross products by a specification of the weights argument in function bam. Our simulations showed, however, that accounting for only the heterogeneous variance does not lead to a substantial improvement in estimation accuracy. The dependencies could also be accounted for by pre-multiplication with the inverse square root of the covariance of the cross products, for which its construction and inversion would become necessary. For our application to the phonetics data, the covariance matrix of the cross products would be a $52,156,566 \times 52,156,566$ dense and unstructured matrix whose construction is not feasible with current technology ($\approx$ 17,500 Terabytes storage space would be required) and we thus do not focus on this extension in the following. However, Eq. (3.8) allows the inclusion of the covariance of the cross products in less complex settings and for (much) smaller data sets and can then result in more efficient estimates. Despite the violations of working assumptions, we achieve good results with our approach in simulations (cp. Section 7). Note that a relevant improvement of the covariance estimation can only be obtained if the working covariance is reasonably well estimated and thus more than one iteration of the algorithm might be necessary. Especially for multiple iterations, our fast smoothing algorithm considerably speeds up the estimation compared to smoothing the entire covariances.

4. Application in functional principal component analysis

An important application of our fast symmetric additive smoothing approach is functional principal component analysis. FPCA is a key tool for dimension reduction in FDA that extracts the dominant modes of variation in the data and provides an explicit variance decomposition. In the following, we briefly outline the four main steps of FPCA for the general FLMM (3.1) using our newly proposed covariance estimation approach. Applying FPCA to Model (3.1) yields parsimonious representations of each random process $\boldsymbol{U}_g(t)$, $g = 1, \ldots, G$, and $E_i(t)$, in bases of eigenfunctions of the respective, previously estimated smooth auto-covariances. In addition, we briefly describe how our covariance smoothing approach can be combined with the general framework of functional additive mixed models (FAMM; Scheipl et al., 2015) allowing for approximate statistical inference for the mean conditional on the FPCA. For a more detailed description, see (Cederbaum et al., 2016).

In the first step, the smooth mean function is commonly estimated based on a working independence assumption (see, e.g., Yao et al., 2005; Di et al., 2009; Greven et al., 2010; Cederbaum et al., 2016). We use penalized splines implemented in R-package mgcv, which can be used to estimate a large variety of covariate and interaction effects in the mean function. Note that the combination with the FAMM framework in the fourth step allows to re-estimate the mean function under a more suitable covariance assumption. For the subsequent steps, the curves are then centered by subtracting the estimated mean from the functional observations.

For the second step, we propose to simultaneously estimate the upper triangles of the covariances $\boldsymbol{K}^{U_g}(t, t')$, $g = 1, \ldots, G$, and $K^E(t, t')$ and the error variance using our novel covariance smoothing approach. The triangular covariance surfaces are then reflected across the diagonal, yielding estimates for the complete covariance surfaces $\boldsymbol{K}^{U_g}(t, t') = \left[K^{U_g}_{ss'}(t, t')\right]_{s,s'=1,\ldots,\rho^{U_g}}$, $g = 1, \ldots, G$, and $K^E(t, t')$. Negative estimated values of σ^2 are set to zero.

In the third step, we use spectral decompositions of the estimated covariance surfaces based on Mercer's Theorem (Mercer, 1909)

$$\boldsymbol{K}^{U_g}(t,t') = \sum_{k=1}^{\infty} \nu_k^{U_g} \boldsymbol{\phi}_k^{U_g}(t) \boldsymbol{\phi}_k^{U_g}(t')^\top, \quad \boldsymbol{K}^{E}(t,t') = \sum_{k=1}^{\infty} \nu_k^{E} \phi_k^{E}(t) \phi_k^{E}(t'),$$

with eigenvalues $\nu_k^{U_g}$, ν_k^E and (vector-valued) eigenfunctions $\boldsymbol{\phi}_k^{U_g}(t) = \left[\phi_{ks}^{U_g}(t)\right]_{s=1,\dots,\rho^{U_g}}$, $\phi_k^E(t)$, respectively. In practice, the covariance surfaces are evaluated on a dense grid $T = \{t_1, \dots, t_D\} \in \mathcal{T}$ of pre-specified length D. We obtain estimated eigenvalues $\widehat{\nu}_k^{U_g}$, $k = 1, \dots, D\rho^{U_g}$, and $\widehat{\nu}_k^E$, $k = 1, \dots, D$, as well as orthonormal eigenfunctions evaluated on T, $\widehat{\boldsymbol{\phi}}_k^{U_g} = \left[\widehat{\phi}_{ks}^{U_g}(t)\right]_{s=1,\dots,\rho^{U_g}, t\in T} \in \mathbb{R}^{D\rho^{U_g}}$ and $\widehat{\boldsymbol{\phi}}_k^E = \left[\widehat{\phi}_k^E(t)\right]_{t\in T} \in \mathbb{R}^D$, of each corresponding covariance operator. The multivariate eigenvectors $\widehat{\boldsymbol{\phi}}_k^{U_g}$ consist of blocks for the respective eigenfunction components. The eigenvectors $\widehat{\boldsymbol{\phi}}_k^{U_g}$, $\widehat{\boldsymbol{\phi}}_k^E$ (and accordingly the eigenvalues) are rescaled to ensure orthonormality with respect to the additive scalar product $\langle (f_1, \dots, f_{\rho^{U_g}}), (g_1, \dots, g_{\rho^{U_g}}) \rangle = \sum_{s=1}^{\rho^{U_g}} \int_{\mathcal{T}} f_s(t) g_s(t) \,\mathrm{d}t$, and with respect to the L^2-scalar product $\langle f, g \rangle = \int_{\mathcal{T}} f(t) g(t) \,\mathrm{d}t$, respectively. For more details on multivariate FPCA, see e.g., (Ramsay and Silverman, 2005). To guarantee positive semi-definiteness of the covariances, which is not ensured by our smoothing approach, negative eigenvalues can be set to zero, which has been shown to improve accuracy in terms of the L^2-norm (Hall et al., 2008) and to work well in practice (Yao et al., 2003). Dimension reduction is achieved by truncating the number of eigenfunctions. We choose the truncation levels based on the proportion of variance explained (see Greven et al., 2010 for an overview) and denote them by N^{U_g}, $g = 1, \dots, G$, and N^E, respectively. The truncated Karhunen–Loève (KL) expansion (Loève, 1946; Karhunen, 1947) allows parsimonious representations of the random processes in truncated bases of the corresponding eigenfunctions

$$\boldsymbol{U}_{gl}(t) \approx \sum_{k=1}^{N^{U_g}} \xi_{lk}^{U_g} \boldsymbol{\phi}_k^{U_g}(t), \quad E_i(t) \approx \sum_{k=1}^{N^E} \xi_{ik}^E \phi_k^E(t), \tag{4.1}$$

with uncorrelated zero mean random basis weights $\xi_{lk}^{U_g}$, $l = 1, \dots, L^{U_g}$, $k = 1, \dots, N^{U_g}$, and ξ_{ik}^E, $i = 1, \dots, n$, $k = 1, \dots, N^E$, with variance $\nu_k^{U_g}$ and ν_k^E, respectively.

In the fourth step, we predict the random basis weights, which give insight into the individual structure of each grouping level. Replacing the random processes in Model (3.1) by their truncated KL expansions in (4.1) allows to approximate the model by a scalar linear mixed model with random effects corresponding to the random basis weights $\xi_{lk}^{U_g}$, ξ_{ik}^E (Di et al., 2009). The basis weights can then be predicted as empirical best linear unbiased predictors by simply plugging-in the estimated eigenfunctions, eigenvalues, and the estimated error variance (Di et al., 2009; Greven et al., 2010; Cederbaum et al., 2016).

Alternatively, we can represent our model as a FAMM using our estimated eigenfunctions and -values in basis expansions of the random processes as proposed by Scheipl et al. (2015). Under the assumption of Gaussian responses, the random basis weights can be predicted together with a re-estimation of the mean function in a mixed model framework. This allows for more efficient mean estimation due to taking the covariance structure into account and for approximate statistical inference conditional on the FPCA, such as pointwise confidence bands for the mean and for covariate effects. For more details on the combination with the FAMM approach and an extensive comparison of the two ways to predict the basis weights, see (Cederbaum et al., 2016).

5. Implementation

We provide freely available software implemented in the R-package sparseFLMM (Cederbaum, 2016). Our implementation is based on the R-package mgcv (Wood, 2011), which allows to add user-defined spline bases and penalties. In this framework, we define a novel class for bivariate smooths estimated subject to our symmetry constraint (2.5) called 'symm' by providing a new constructor method function smooth.construct.symm.smooth.spec and a corresponding predictor method function for the estimation of smooth surfaces in additive models. The class can be applied to any bivariate smooth term in a gam-formula. It is not restricted to symmetric data although we here consider symmetric data in the form of cross products in the smoothing of the covariance. It can be applied to (possibly noisy) data sampled on a regular grid as well as to irregularly or sparsely sampled data. As a modular component, our constructor can be applied separately to the auto-covariances of independent functional random effects in an FLMM. The case of correlated functional random effects, such as correlated functional random intercepts and slopes, is currently not covered in the implementation. In our application, we show how the constructor can be applied to complex designs on the basis of the FLMM with crossed fRIs as in (3.7). One main advantage of using standard software is that it allows for flexible extensions. The current implementation in the R-package sparseFLMM is based on tensor product B-splines with difference penalties (Eilers and Marx, 2003), but extensions to other bivariate bases that are symmetric across the diagonal and other penalties are possible.

The spline degree and the number of basis functions can be chosen. The user currently has the choice between two different quadratic penalties. One can either use the Kronecker sum penalty $\text{pen}(\lambda) = \lambda\boldsymbol{\theta}^\top [(\boldsymbol{S}_t \otimes \boldsymbol{I}_F) + (\boldsymbol{I}_F \otimes \boldsymbol{S}_{t'})]\boldsymbol{\theta}$ or alternatively a Kronecker product penalty of the form $\text{pen}(\lambda) = \lambda\boldsymbol{\theta}^\top (\boldsymbol{S}_t \otimes \boldsymbol{S}_{t'})\boldsymbol{\theta} = \lambda\boldsymbol{\theta}^\top [(\boldsymbol{S}_t \otimes \boldsymbol{I}_F) \cdot (\boldsymbol{I}_F \otimes \boldsymbol{S}_{t'})]\boldsymbol{\theta}$, where $\boldsymbol{S}_t = \boldsymbol{S}_{t'}$ denote the marginal penalty matrices and $\boldsymbol{\theta}$ is the coefficient vector. Note that both penalties are isotropic, reflecting the symmetry of the surface. The main difference between the two penalty matrices is that the null space of the latter is larger, more likely leading to wigglier estimates. Other possible penalties could be added by the user.

To speed up estimation, function bam can compute the computationally expensive steps in parallel on multiple cores.

In addition to the implementation of our constructor, the R-package sparseFLMM includes an implementation for the FPCA based on our (additive) covariance smoothing approach for three special cases of the FLMM (Model (2.1), a model with an fRI and a smooth error curve, and the model with two crossed fRIs and a smooth error curve as in Section 3.3).

6. Applications

We demonstrate the practical relevance of our approach in two distinct applications. We consider a standard data set consisting of sparse longitudinal observations as well as functional data with a complex correlation structure and different grids between curves.

6.1. CD4 cell count data

In order to compare our approach to that of Xiao et al. (2017) for the common special case of longitudinal data, we apply FPCA to analyze the CD4 cell count trajectories in HIV positive individuals which are available in R-package refund (Huang et al., 2016). As our focus here is on the more complex case of correlated functional data, to which the approach of Xiao et al. (2017) does not apply, the application to the CD4 cell count data is given in Appendix C.

6.2. Phonetics data

We apply FPCA based on our covariance smoother to acoustic signal data with a crossed correlation structure. We show the increase of the computational efficiency compared to the approach of Cederbaum et al. (2016), for a case where their approach is applicable. To the best of our knowledge, their smooth method of moments approach is the only competitor for covariance smoothing of irregularly observed correlated curves with a crossed design structure. No alternative approach is available for general models (3.1), where our approach is the first available for additive covariance smoothing.

In phonetic research, the term consonant assimilation refers to the phenomenon that the articulation of two consonants becomes phonetically more alike when they appear subsequently in fluent speech. Consonant assimilation is accompanied by a complex interaction of language-specific, perceptual and articulatory factors which makes it an important topic in speech production research. The data we consider are part of a large study which was conducted by Pouplier and Hoole (2016) in order to investigate among others the assimilation of the consonants /s/ and /sh/ as a function of their order (/s#sh/ versus /sh#s/, where # denotes a word boundary), syllable stress and vowel context in the German language. The same data were previously analyzed by Cederbaum et al. (2016). Pouplier and Hoole (2016) recorded the audio signals for nine native speakers which repeated the same sixteen target words each five times. The target words consisted of (semantically nonsensical) bisyllabic noun–noun compound words with abutting consonants /s/ and /sh/ in either order, e.g., 'Calla**s**-**Sch**immel' and 'Gula**sch**-**S**impel', and with either stressed or unstressed syllables and varying vowel combinations. The time interval during the duration of the two consonants of interest was cut out manually by the phoneticians and standardized to a [0,1] interval in which the recorded acoustic signals were summarized in a functional index over time. The $n = 707$ index curves (shown in Fig. 1) take values between $+1$ and -1, with positive [negative] values indicating proximity of the signal to a reference signal for the first [second] consonant of the target word, respectively. To illustrate the effect of consonant assimilation, two acoustic signals are highlighted in Fig. 1. The curve without or with little assimilation shows a clear transition from a strong positive to a strong negative value, whereas the curve with strong assimilation is quite flat and mostly takes negative values. The curves differ in the number and location of the observation points, ranging from 22 to 57 with a median of 34 points per curve. For a more detailed description of the data (including pre-processing steps), see (Pouplier and Hoole, 2016; Cederbaum et al., 2016).

We fit an FLMM with crossed fRIs as described in Section 3.3 and previously considered in Cederbaum et al. (2016) to account for the repeated measurements of both speakers and target words. The mean function $\mu(t, \boldsymbol{x}_i)$ includes effects and interaction effects of the dummy-coded covariates consonant order, syllable stress, and vowel context, smoothly varying over time. As the focus here is on the smooth auto-covariances of the functional random effects, $K^B(t, t')$, $K^C(t, t')$, and $K^E(t, t')$, we refer to (Cederbaum et al., 2016) for more details on and interpretations of the covariate effects. For each auto-covariance, we use cubic marginal B-spline bases with marginal third order difference penalty matrices and use the Kronecker sum penalty for bivariate smoothing (cp. Section 5). We compare the results from our novel symmetric bivariate smoother (denoted by TRI-CONSTR and by TRI-CONSTR-W with weights of 0.5 for the cross products on the diagonal, cp. Section 2.2) with the results we obtain using the smoothing approach of Cederbaum et al. (2016) (denoted by WHOLE). The latter does not exploit the symmetry of the estimated surface and is equivalent to TRI-CONSTR-W except for the estimation of the smoothing parameter and numerical differences. To highlight the need for a symmetry constraint when only the upper

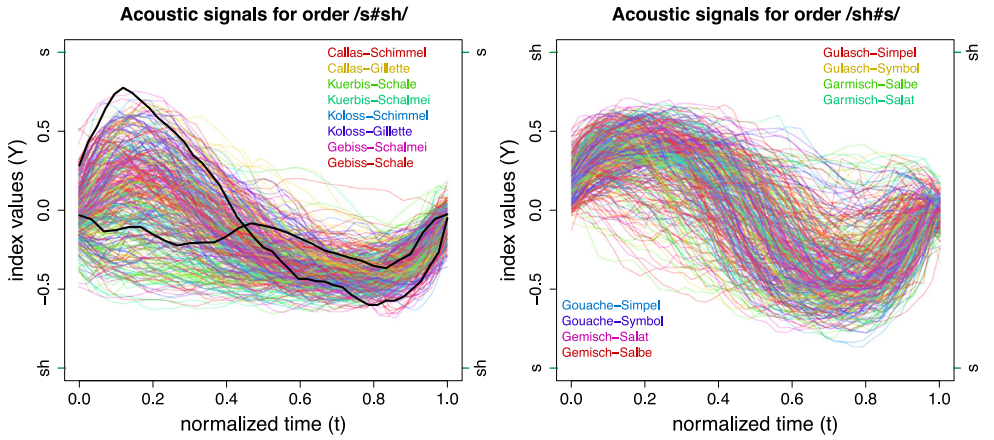


Fig. 1. Acoustic signal curves of the phonetics data over time (cf. Cederbaum et al., 2016). Left [right]: Signal curves of consonant order /s#sh/ [sh#s] colored by target word. Example curves with no/little assimilation and with strong assimilation highlighted (black).

triangle is considered, we further compare with the results obtained by estimating the upper triangle without a symmetry constraint (denoted by TRI) which does not guarantee smoothness across the diagonal (cp. Section 2.2). The estimated auto-covariances are evaluated on a pre-specified grid of length $D = 100$. During the FPCA, we choose the truncation levels based on a pre-specified proportion of explained variance of $L = 0.95$, yielding two [four] eigenfunctions for the auto-covariance of the fRI for speakers, $K^B(t, t')$, and three [twelve] eigenfunctions for the auto-covariance of the smooth error, $K^E(t, t')$, for our two approaches and WHOLE [for TRI]. For all four smoothing methods, no eigenfunction is chosen for the fRI for target words, which is due to the high number of covariates that describe the target words sufficiently (cf. Cederbaum et al., 2016). Fig. 2 shows the estimated surfaces and contours of the auto-covariance of the fRI for speakers, reconstructed after truncation from the estimated eigenvalues and eigenfunctions, which are shown in the bottom of the figure (black). As a sensitivity analysis, we additionally depict jackknife estimates of the eigenfunctions with each of the nine speakers left out once (gray lines, bottom row). The variability over jackknife samples is similar for all compared approaches. However, while the same number of FPCs is chosen in all jackknife samples for TRI-CONSTR, TRI-CONSTR-W, and WHOLE, jackknife results for TRI are more sensitive. In two out of nine jackknife samples, TRI selects more FPCs than for the whole data set.

We can see from Fig. 2 that the estimated covariance $\widehat{K}^B(t, t')$ based on our symmetric smoother (TRI-CONSTR-W) is very similar to the one obtained by using all cross products (WHOLE). Moreover, it shows that for this application, the weights for the diagonal cross products do not make a great difference. As expected, we observe wigglier estimates especially on the diagonal for TRI, for which the error variance is estimated to be zero. This is also reflected in the wiggliness and higher number of chosen eigenfunctions for TRI. Similar results can be found for $K^E(t, t')$. These are given in Appendix C, where additional estimation details and results including the complete variance decompositions are provided. For all four approaches, the first eigenfunction in Fig. 2 (solid line) corresponds to the discrimination of the speaker between the first and the second consonant and the second eigenfunction (dashed line) mainly leads to a vertical shift of the signal curves. Accounting for the symmetry of the covariances leads to a considerable reduction of computation times. TRI-CONSTR and TRI-CONSTR-W have the shortest computation times for smoothing the three auto-covariances, using five kernels in parallel, amounting to 24.51 and 25.82 min, respectively. Smoothing the auto-covariances using WHOLE takes more than twice as long (55.76 min) and using TRI still amounts to 32.95 min, which partly results from the fact that more spline coefficients have to be estimated. In addition, WHOLE and TRI require the estimation of two (instead of one) smoothing parameters for each auto-covariance using the Kronecker sum penalty implemented in R-package mgcv of the form $\text{pen}(\lambda) = \lambda_t \boldsymbol{\theta}^\top (\boldsymbol{S}_t \otimes \boldsymbol{I}_F)\, \boldsymbol{\theta} + \lambda_{t'} \boldsymbol{\theta}^\top (\boldsymbol{I}_F \otimes \boldsymbol{S}_{t'})\, \boldsymbol{\theta}$.

7. Simulations

7.1. Simulation designs

To investigate the performance of our covariance smoothing approach, we conduct an extensive simulation study based on two different data generating processes. For the first scenario (Scenario 1), we simulate data consisting of $n = 100$ independent curves from Model (2.1) with $\mu(t) = \sin(t) + t$. The data for the second scenario (Scenario 2) are generated from an FLMM with crossed fRIs as in Section 3.3, such that they mimic the irregularly observed phonetics data. Note that no covariate mean effects are included such that we additionally obtain one eigenfunction for the fRI for target words and thus really have crossed fRIs. Additional generation details for this scenario are given in Appendix D. We compare the performance of our fast (additive) symmetric covariance smoother (TRI-CONSTR, TRI-CONSTR-W with weights of 0.5 for the diagonal cross products) as basis for FPCA with the performance of the smoothing approach proposed by Cederbaum et al. (2016)

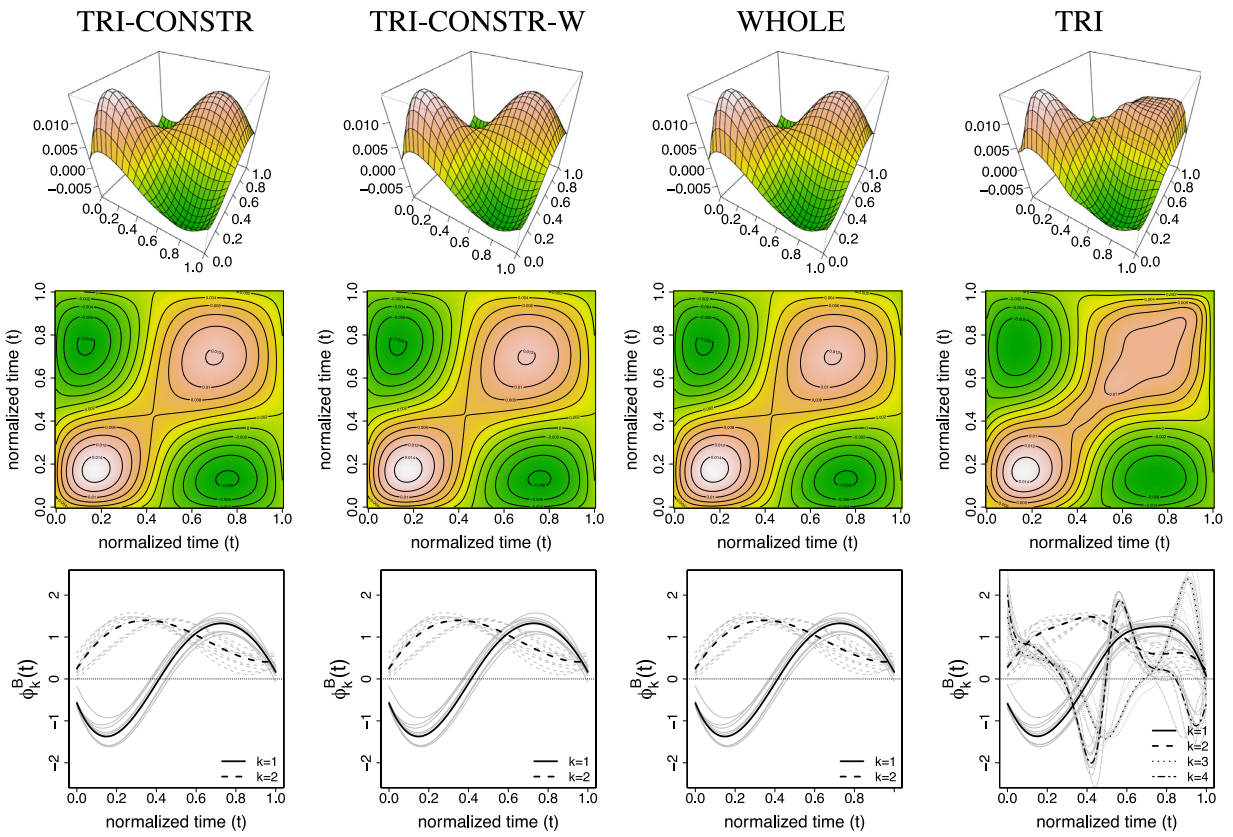


Fig. 2. Results for the fRI for speakers using the four smoothing methods. Top row: estimated covariance surfaces. Middle row: contours of the estimated covariance surfaces. Bottom row: estimated eigenfunctions $\widehat{\phi}_k^B(t)$ based on the entire data set (in black) and jackknife estimates with one of the speakers left out in turn (in gray).

(WHOLE). To evaluate the need of the symmetry constraint when only the upper triangle is estimated, we further compare with the results obtained without posing a symmetry constraint (TRI). For Scenario 1, we additionally provide a comparison with the smoothing approach of Xiao et al. (2017) (denoted by FACE), implemented in function face.sparse in R-package face (Xiao et al., 2016). FACE does, however, not apply to additive covariance smoothing needed for correlated curves. Note that Xiao et al. (2017) compare their symmetric covariance smoother to a number of other approaches that are all restricted to independent curves. They show that their approach is superior or comparable in terms of median integrated squared errors and inter quartile ranges (IQR) to the approach implemented in function fpca.sc (Di et al., 2009) in R-package refund based on bivariate B-splines with a difference penalty and to a self-coded variant based on thin plate regression splines for covariance smoothing. Moreover, they demonstrate their supremacy over the geometric likelihood approach of Peng and Paul (2009) and the local polynomial approach proposed in Yao et al. (2003). We thus do not include these alternatives here.

Based on Scenario 1, we investigate the sensitivity of the estimates to varying model complexity in terms of the complexity of the underlying eigenfunctions, signal to noise levels as functions of error variances and eigenvalues, and degree of sparseness. We consider all possible combinations of

1. simple eigenfunctions: $\{\phi_1(t) = 1, \phi_2(t) = \sqrt{3}(2t-1)\}$, complex eigenfunctions $\{\phi_1(t) = \sin(2\pi t), \phi_2(t) = \cos(2\pi t)\}$
2. error variance: $\sigma^2 = 0.5, \sigma^2 = 0.05$
3. eigenvalues: $\{\nu_1 = 0.15, \nu_2 = 0.075\}, \{\nu_1 = 2, \nu_2 = 1\}$
4. number of observation points: drawn from uniform distributions $\mathcal{U}[40, 60]$.

For the simple eigenfunctions, we additionally consider a sparse setting, in which the number of observation points is drawn from the uniform distribution $\mathcal{U}[3, 10]$. For the complex eigenfunctions and eigenvalues $\nu_1 = 2$, $\nu_2 = 1$, we additionally reduce the value of the error variance to $\sigma^2 = 0.01$. For all settings, we generate 200 data sets. The random basis weights are centered and decorrelated such that the weights empirically have zero mean and a correlation of zero (see Cederbaum et al., 2016, for a discussion).

For the estimation in Scenario 1, we use ten cubic B-splines each for the estimation of the mean function and as marginal bases for the auto-covariances. For the estimation in Scenario 2, we use eight and five cubic B-splines for the estimation

of the mean function and as marginal bases for the auto-covariances, respectively. We use Kronecker sum penalties (cp. Section 5) of marginal third order difference penalties for bivariate smoothing. Estimation of the smoothing parameter is based on REML, except for FACE, which uses leave-one-curve-out cross-validation. We use equidistant knots in function face.sparse instead of the default (quantile based knots) which would require an adapted penalty that is not implemented. The arguments that determine the smoothing parameter search in function face.sparse are left at their defaults. As function face.sparse does not allow to specify a fixed truncation level, we choose the number of eigenfunctions based on a pre-specified proportion of explained variance of $L = 0.95$. Note that we use the proportion of explained variance in the functional observations, whereas (Xiao et al., 2016) use that in $E_i(t)$. In order to be able to differentiate the error incurred by the truncation of the covariance surface to a few leading FPCs from the pure covariance surface estimation error, we pre-specify the correct truncation lags for Scenario 2, for which no comparison with FACE is possible anyway.

7.2. Simulation results

We present and discuss the results of both simulation scenarios. For Scenario 1, we focus our presentation of the results on the setting with complex eigenfunctions, an error variance of $\sigma^2 = 0.05$ and eigenvalues of size $\nu_1 = 2$, $\nu_2 = 1$ (denoted as Setting 1). As a measure of goodness of fit, we use root relative mean squared errors (rrMSEs) of the form $\sqrt{(\text{true} - \text{estimated})^2/\text{true}^2}$. The complete results for all settings and the specific forms of the rrMSE for all model components, are given in Appendix D.

Fig. 3 depicts boxplots of the rrMSEs for 200 simulation runs for Setting 1. For each model component, it shows the boxplots for the compared smoothing methods and, in addition, for a modified version of FACE (denoted by FACE-STEP-1), in which the covariance of the cross products is not accounted for and thus only the first step of the three-step procedure is performed. FACE-STEP-1 is added to evaluate the effect of accounting for the covariance of the cross products (cf. Section 3.4). It shows that all components, except for the error variance, are estimated very well for our approach (TRI-CONSTR, TRI-CONSTR-W). The weights for the diagonal cross products do not have a great influence. Moreover, our approach yields similar rrMSEs as WHOLE. We also obtain similar results for TRI for most components, except for the error variance for which TRI has a higher median rrMSE. FACE yields a more than 2.7 times higher median rrMSE for the auto-covariance $K^E(t, t')$ compared to our approach and consequently also worse results for the eigenvalues ν_k^E, $k = 1, 2$. It yields smaller median rrMSEs for the eigenfunctions ϕ_k^E, $k = 1, 2$, but the IQR is larger and more outliers occur than in all other approaches. The estimation of the error variance profits most from accounting for the covariance of the cross products, which is reflected in a much lower median rrMSE for FACE than for the other methods. This also leads to lower median rrMSEs for the random basis weights, ξ_1^E and ξ_2^E, which depend on the error variance and consequently also to lower median rrMSEs for the reconstructed processes $E_i(t)$ and $Y_i(t)$. For all components, however, FACE yields a high number of outliers that range above the maximal rrMSEs obtained for the other methods. It is noticeable that the estimation of the auto-covariance is considerably better (in terms of rrMSE) for FACE when the covariance of the cross products is not accounted for. Moreover, we see that FACE-STEP-1 yields roughly similar results to our approach and WHOLE. For TRI-CONSTR, TRI-CONSTR-W and WHOLE, the truncation level is correctly estimated to be two in all 200 simulation runs. A higher number of eigenfunctions (three to four) is chosen for TRI in eight simulation runs of this setting, which is consistent with our results in the application to the phonetics data. FACE and FACE-STEP-1 choose more than two (three to five) eigenfunctions in 190 and 198 simulation runs, respectively.

To sum up the results for the other ten settings of Scenario 1, we can say that over all settings and components TRI-CONSTR, TRI-CONSTR-W and WHOLE yield similar rrMSEs with a tendency to a supremacy of TRI-CONSTR, especially in the sparse settings. TRI yields similar to worse results compared to our method and WHOLE. Especially for the error variance it yields up to 82% higher median rrMSEs (in one of the sparse settings). The estimation quality of all methods differs between the dense and the sparse settings. Our approach yields relatively similar results within the dense settings and higher rrMSEs in the sparse settings. Moreover, TRI-CONSTR and TRI-CONSTR-W tend to perform better in the settings with complex eigenfunctions and favor smaller error variances (except for the estimation quality of the error variance itself). In contrast, FACE tends to perform better in the settings with simple eigenfunctions and favors larger error variances. Our approach and WHOLE always select the correct truncation level, except in the sparse settings, where for some simulation runs more eigenfunctions are selected. TRI tends to select more eigenfunctions. For all settings, FACE and FACE-STEP-1 have simulation runs in which more than two eigenfunctions are selected.

In contrast to FACE and FACE-STEP-1, our approach is directly extendable to smoothing multiple auto-covariances simultaneously, which is required in Scenario 2. Fig. 4 depicts the rrMSEs for the auto-covariances, the error variance and curves $Y_i(t)$ for Scenario 2. It shows that the three auto-covariances are estimated equally well for the four compared smoothing methods. For the error variance, however, TRI performs worse with a 32.5% higher median rrMSE and a larger IQR, which also results in slightly higher rrMSEs for the reconstructed curves. Over all, our methods perform very well considering the small number of levels for the fRIs B (9 levels) and C (16 levels). For all model components, TRI-CONSTR and TRI-CONSTR-W yield similar rrMSEs.

7.3. Computational efficiency

Fig. 5 shows computation times on a 64 Bit Linux platform with 660 Gb of RAM memory for the two settings discussed above. Our approach (with and without weights for the diagonal cross products) greatly speeds up the computation

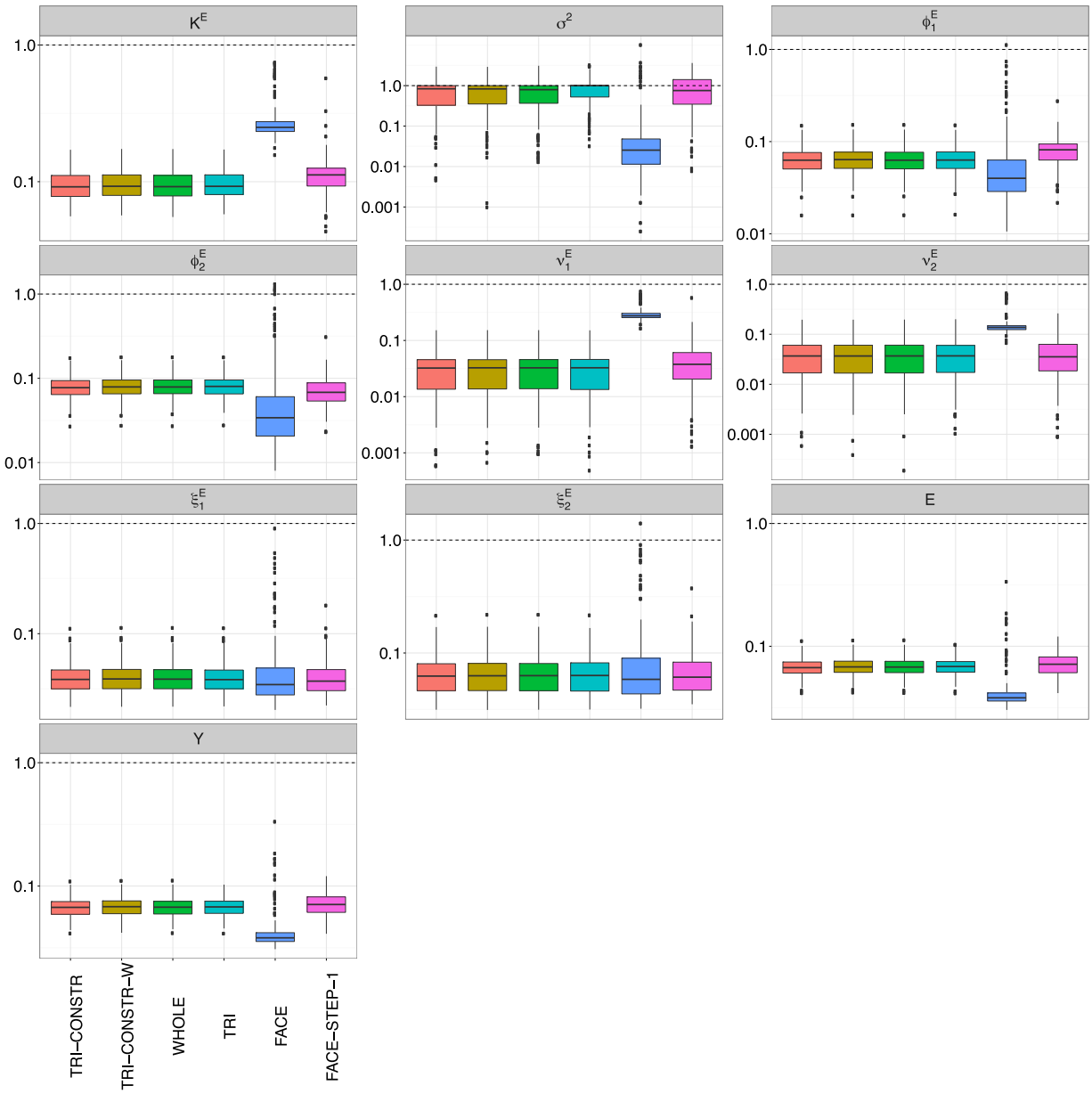


Fig. 3. Boxplots of the rrMSEs (log10 scale at y-axis) for the scenario with independent curves (Scenario 1) for the smoothing methods being compared. Top row: rrMSEs for auto-covariance $K^E(t, t')$, error variance σ^2, and the first eigenfunction $\phi_1^E(t)$. Second row: rrMSEs for the second eigenfunction $\phi_2^E(t)$ and eigenvalues ν_1^E, ν_2^E. Third row: rrMSEs for the random basis weights ξ_1^E, ξ_2^E and process $E_i(t)$. Bottom row: rrMSEs for curves $Y_i(t)$.

compared to WHOLE and also to TRI, especially in the scenario with crossed fRIs (right figure), where the computation times are longer and thus matter more. Note that in addition to accounting for the symmetry of covariances, we reduce the number of smoothing parameters to be estimated compared to WHOLE and TRI, for which two smoothing parameters are estimated for each auto-covariance using the Kronecker sum penalty implemented in R-package mgcv. In the scenario with independent curves (left figure), FACE is considerably slower than the other methods. This applies to all except the sparse settings, in which the computation times are extremely short anyway (max. 15 s). Note that variation for FACE and FACE-STEP-1 is considerably smaller as the smoothing parameter is chosen based on a fixed grid.

7.4. Summary

Overall, one can conclude from our simulations that the proposed symmetric smoothing approach performs well both in the simple scenario as well as in the scenario with crossed fRIs. In the regarded 12 simulation settings, the novel symmetric

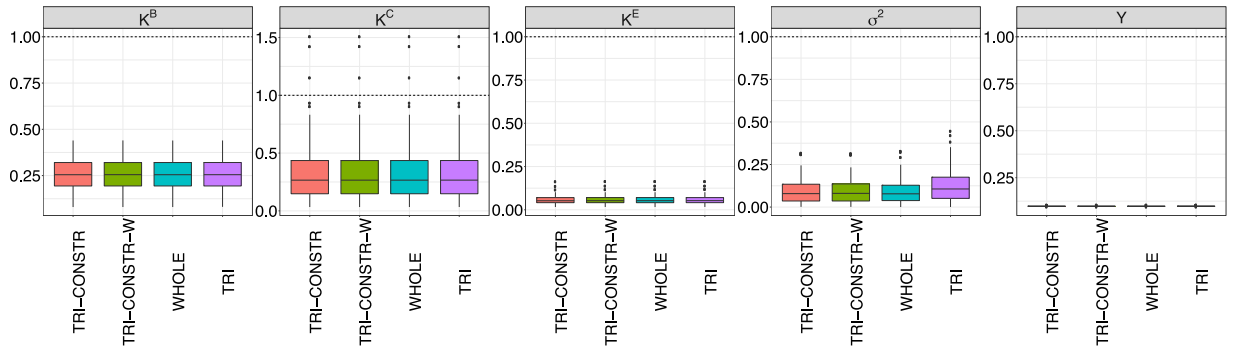


Fig. 4. Boxplots of the rrMSEs for the scenario with crossed fRIs (Scenario 2) for the smoothing methods being compared. Shown are boxplots of the rrMSEs for the three auto-covariances $K^B(t, t')$, $K^C(t, t')$, and $K^E(t, t')$ as well as for the error variance σ^2 and the curves $Y_i(t)$.

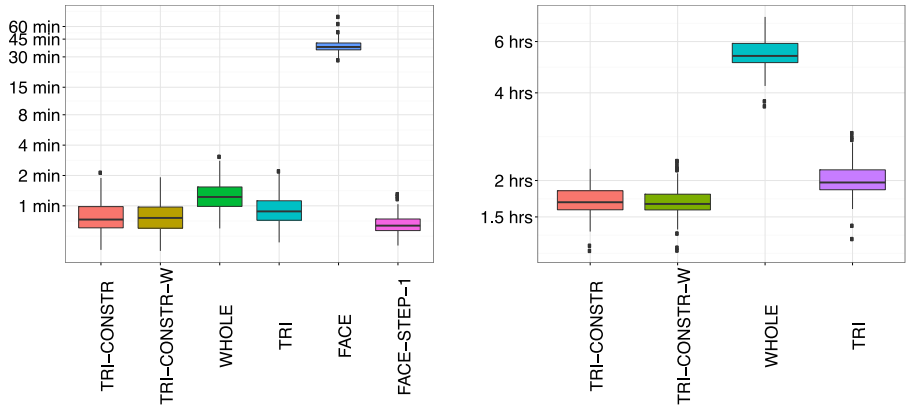


Fig. 5. Computation times (log10 scale at y-axis) for 200 simulation runs for all compared methods. Left: computation times for Setting 1 of the scenario with independent curves. Right: computation times for the scenario with crossed fRIs.

smoother yields similar results with and without weights for the cross products on the diagonal. It shows that the results are also similar to those of the approach of Cederbaum et al. (2016) that, however, requires considerably longer computation times. As expected, the proposed symmetric smoothing approach performs better than the estimation of the upper triangle without a symmetry constraint, most notably for the estimation of the error variance. For the scenario with independent curves, in which a comparison with the approach of Xiao et al. (2017) is possible, neither is clearly preferable for all estimated components in all settings. In addition to the computational advantages of the proposed symmetric smoother, it also has much fewer outliers than FACE and more reliably selects the correct number of FPCs. Most importantly, the proposed approach – in contrast to FACE – allows to smooth multiple auto-covariances simultaneously.

8. Discussion and outlook

We have introduced a fast bivariate smoothing approach for symmetric surfaces which applies to a broad range of data situations. We focus on its application to estimate covariance functions in longitudinal data as well as multiple additive covariance functions in functional data with very general correlation structures. Our smoother can handle (possibly noisy) data sampled on a common, dense grid as well as irregularly or sparsely observed data, which are frequently encountered in practice. It extends the smooth method of moments estimator of Cederbaum et al. (2016) to more general correlation structures and additionally takes advantage of the symmetry of the sample covariances, which leads to considerably faster estimation requiring less memory. The covariance functions are estimated as unknown, smooth functions in a bivariate additive varying coefficient model for the empirical covariances (cross products). A symmetry constraint additionally ensures smoothness of the estimated covariance surfaces across the diagonal and further reduces computational costs. We show how our smoother can be applied as basis for FPCA, a key tool for dimension reduction in FDA, and demonstrate its practical relevance in a longitudinal data application and in an application to complexly correlated functional phonetics data.

We provide fully documented open-source software implementing our approach in the R-package sparseFLMM, which builds on the established R-package mgcv (Cederbaum, 2016; Wood, 2011) allowing for flexible extensions. Within this framework, we provide a novel constructor function for the estimation of smooth surfaces in additive models subject to the

symmetry constraint. Our constructor can be applied as a modular component to general bivariate symmetric smoothing problems.

Simulation experiments (in Section 7) show that the proposed method recovers the true functions very well and yields similar results as the estimation approach of Cederbaum et al. (2016) while considerably speeding up the estimation and extending the range of possible model structures.

This work opens up a number of interesting directions for future research. A first direction concerns the working assumptions of independent and homoscedastic cross products that are implicitly made by using a quadratic loss function in the estimation of the covariances in a bivariate additive varying coefficient model (cp. Section 3.4). It would be interesting to investigate whether a suitable loss function for the cross products could be derived. Under the assumption of Gaussian responses, the cross products follow a product normal distribution, for whose probability density function (Nadarajah and Pogány, 2016) recently derived a closed-form expression based on the modified Bessel function. A second direction concerns the positive semi-definiteness of the covariance operator, which is not ensured in our approach and in most existing covariance smoothing approaches (e.g., Yao et al., 2003; Hall et al., 2008; Di et al., 2009; Greven et al., 2010). Although (Hall et al., 2008) show that setting negative eigenvalues to zero improves the estimation quality and (Yao et al., 2003) demonstrate that this works well in practice, it could be desirable to ensure positive semi-definiteness in the estimation. Wu and Pourahmadi (2003) estimate a positive semi-definite covariance based on an auto-regressive model with regression coefficients corresponding to the components of a modified Cholesky decomposition of the covariance. The approach of Peng and Paul (2009) also ensures positive semi-definiteness. It remains an open question, however, how these approaches could be extended to smoothing multiple additive covariances for functional data with complex correlation structures. Another possible path for future investigation would be the extension of our models to allow for more complex noise terms. This extension requires careful, extensive analyses in order to ensure identifiability. For much simpler models, (Descary and Panaretos, 2016) develop identifiability conditions for the separation of smooth and rough variation (corresponding to $E_i(t_{ij})$ and $\varepsilon_i(t_{ij})$ in our notation). Whether and how these conditions can be extended to the error terms of the very general functional linear mixed models considered in this paper remain an open question.

Acknowledgments

The authors thank Marianne Pouplier and Phil Hoole for supplying and explaining the phonetics data. The authors also thank the referees and the editor for their useful comments. Funding: This work was supported by the German Research Foundation [Emmy Noether grant number GR 3793/1-1].

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.csda.2017.11.002.

References

Besse, P., Ramsay, J.O., 1986. Principal components analysis of sampled functions. Psychometrika 51 (2), 285–311.
Cederbaum, J., 2016. sparseFLMM: Functional linear mixed models for irregularly or sparsely sampled data, R package version 0.1.0. Available at https://CRAN.R-project.org/package=sparseFLMM.
Cederbaum, J., Pouplier, M., Hoole, P., Greven, S., 2016. Functional linear mixed models for irregularly or sparsely sampled data. Stat. Model. 16 (1), 67–88.
Descary, M.-H., Panaretos, V.M., 2016. Functional Data Analysis by Matrix Completion, arXiv preprint: arXiv:1609.00834v1. (in press).
Di, C.-Z., Crainiceanu, C.M., Caffo, B.S., Punjabi, N.M., 2009. Multilevel functional principal component analysis. Ann. Appl. Stat. 3 (1), 458–488.
Di, C.-Z., Crainiceanu, C.M., Jank, W.S., 2014. Multilevel sparse functional principal component analysis. Statistics 3 (1), 126–143.
Eilers, P.H., Marx, B.D., 2003. Multivariate calibration with temperature interaction using two-dimensional penalized signal regression. Chemometr. Intell. Lab. Syst. 66 (2), 159–174.
Fan, J., Huang, T., Li, R., 2007. Analysis of longitudinal data with semiparametric estimation of covariance function. J. Amer. Statist. Assoc. 102 (478), 632–641.
Faraway, J.J., 1997. Regression analysis for a functional response. Technometrics 39 (3), 254–261.
Greven, S., Crainiceanu, C.M., Caffo, B., Reich, D., 2010. Longitudinal functional principal component analysis. Electron. J. Stat. 4, 1022–1054.
Hall, P., Müller, H., Yao, F., 2008. Modelling sparse generalized longitudinal observations with latent Gaussian processes. J. Roy. Statist. Soc.: Ser. B (Statist. Methodol.) 70 (4), 703–723.
Huang, L., Scheipl, F., Goldsmith, J., Gellar, J., Harezlak, J., McLean, M.W., Swihart, B., Xiao, L., Crainiceanu, C., Reiss, P., 2016. refund: Regression with Functional Data, R package version 0.1-14. Available at https://CRAN.R-project.org/package=refund.
Isserlis, L., 1918. On a formula for the product-moment coefficient of any order of a normal frequency distribution in any number of variables. Biometrika 12 (1–2), 134–139. http://biomet.oxfordjournals.org/content/12/1-2/134.
James, G.M., Hastie, T.J., 2001. Functional linear discriminant analysis for irregularly sampled curves. J. Roy. Statist. Soc.: Ser. B (Statist. Methodol.) 63 (3), 533–550.
James, G.M., Hastie, T.J., Sugar, C.A., 2000. Principal component models for sparse functional data. Biometrika 87 (3), 587–602.
Karhunen, K., 1947. Über lineare Methoden in der Wahrscheinlichkeitsrechnung. Ann. Acad. Sci. Fennicae. Ser. A.1, Math.-Phys. 37, 1–79.
Kauermann, G., Wegener, M., 2011. Functional variance estimation using penalized splines with principal component analysis. Stat. Comput. 21 (2), 159–171.
Krivobokova, T., Kauermann, G., 2007. A note on penalized spline smoothing with correlated errors. J. Amer. Statist. Assoc. 102 (480), 1328–1337.
Leurgans, S.E., Moyeed, R.A., Silverman, B.W., 1993. Canonical correlation analysis when the data are curves. J. Roy. Statist. Soc. Ser. B (Statist. Methodol.) 55 (3), 725–740.
Loève, M., 1946. Fonctions aléatoires du second ordre. La Rev. Sci. 84, 195–206.

Mercer, J., 1909. Functions of positive and negative type, and their connection with the theory of integral equations. Philos. Trans. Roy. Soc. Lond. Ser. A (Statist. Soc.) 209 (441–458), 415–446.
Morris, J.S., 2015. Functional Regression. Annu. Rev. Statist. Appl. 2 (1), 321–359.
Nadarajah, S., Pogány, T.K., 2016. On the distribution of the product of correlated normal random variables. C. R. Math. 354 (2), 201–204.
Patterson, H.D., Thompson, R., 1971. Recovery of inter-block information when block sizes are unequal. Biometrika 58 (3), 545–554.
Peng, J., Paul, D., 2009. A geometric approach to maximum likelihood estimation of the functional principal components from sparse longitudinal data. J. Comput. Graph. Statist. 18 (4), 995–1015.
Pinheiro, J.C., Bates, D.M., 2000. Mixed-Effects Models in S and S-PLUS. In: Statistics and Computing, Springer, New York.
Pouplier, M., Hoole, P., 2016. Articulatory and Acoustic Characteristics of German Fricative Clusters. Phonetica 73 (1), 52–78.
Ramsay, J.O., Silverman, B.W., 2005. Functional Data Analysis, second ed.. In: Springer Series in Statistics, Springer, New York.
R Core Team, , 2016. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria, URL https://www.R-project.org/.
Reiss, P.T., Huang, L., Mennes, M., 2010. Fast Function-on-Scalar Regression with Penalized Basis Expansions. Int. J. Biostat. 6 (1), 1–30.
Reiss, P.T., Ogden, R.T., 2009. Smoothing parameter selection for a class of semiparametric linear models. J. Roy. Statist. Soc.: Ser. B (Statist. Methodol.) 71 (2), 505–523.
Scheipl, F., Staicu, A.-M., Greven, S., 2015. Functional Additive Mixed Models. J. Comput. Graph. Statist. 24 (2), 477–501.
Shou, H., Zipunnikov, V., Crainiceanu, C.M., Greven, S., 2015. Structured functional principal component analysis. Biometrics 71 (1), 247–257.
Staniswalis, J., Lee, J., 1998. Nonparametric Regression Analysis of Longitudinal Data. J. Amer. Statist. Assoc. 93 (444), 1403–1404.
Tai, W., 2009. Regularized Estimation of Covariance Matrices for Longitudinal Data Through Smoothing and Shrinkage. Columbia University, (Ph.D. thesis).
Wood, S.N., 2011. Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models. J. R. Stat. Soc. Ser. B Stat. Methodol. 73 (1), 3–36.
Wu, W.B., Pourahmadi, M., 2003. Nonparametric estimation of large covariance matrices of longitudinal data. Biometrika 90 (4), 831–844.
Xiao, L., Li, C., Checkley, W., Crainiceanu, C., 2016. A face: Fast Covariance Estimation for Sparse Functional Data, R package version 0.1-1. Available at https://CRAN.R-project.org/package=face.
Xiao, L., Li, C., Checkley, W., Crainiceanu, C.M., 2017. Fast covariance estimation for sparse functional data. Stat. Comput.. http://dx.doi.org/10.1007/s11222-017-9744-8. (in press).
Xiao, L., Zipunnikov, V., Ruppert, D., Crainiceanu, C.M., 2016b. Fast covariance estimation for high-dimensional functional data. Stat. Comput. 26 (1), 409–421.
Yao, F., Müller, H.-G., Clifford, A.J., Dueker, S.R., Follett, J., Lin, Y., Buchholz, B.A., Vogel, J.S., 2003. Shrinkage estimation for functional principal component scores with application to the population kinetics of plasma folate. Biometrics 59 (3), 676–685.
Yao, F., Müller, H.-G., Wang, J.-L., 2005. Functional data analysis for sparse longitudinal data. J. Amer. Statist. Assoc. 100 (470), 577–590.